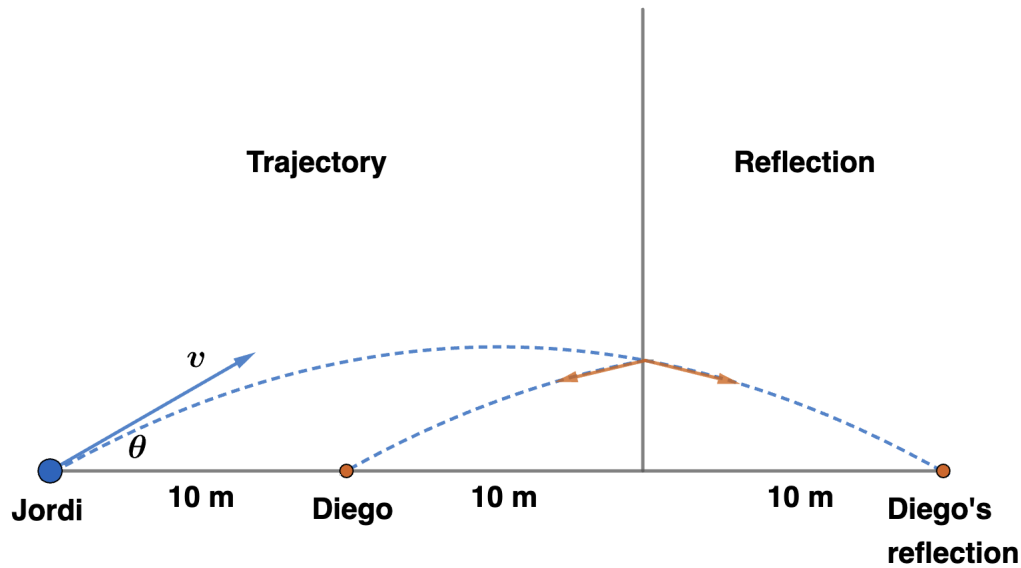


2013 F=ma Exam: Problem 2

Kevin S. Huang



When the ball hits the wall, the horizontal component of the velocity flips direction. From that point onwards, the path taken is the reflection of the trajectory the ball would have taken without the wall.

Thus, for the ball to reach Diego with the wall present, we must reach Diego's reflection when the wall is absent. Jordi and Diego's reflection are separated by $R = 20 \text{ m} + 10 \text{ m} = 30 \text{ m}$ so by the range equation,

$$R = \frac{v^2 \sin(2\theta)}{g}$$

Solving for v ,

$$v = \sqrt{\frac{gR}{\sin(2\theta)}} = \sqrt{\frac{(10 \text{ m/s}^2)(30 \text{ m})}{\sin(2 \cdot 30^\circ)}} = 19 \text{ m/s}$$

so the answer is C.